4.7 ZHMS_R07 — Billing & Revenue Analysis • Main SELECT: zhms_bill_hdr JOIN zhms_bill_item; selection on PATIENT_ID, PAY_STATUS, BILL_DATE range. • ALV Footer: Total Collection (SUM of TOTAL_AMOUNT where PAYMENT_STATUS = 'P'), Outstanding Amount (SUM where PAYMENT_STATUS NE 'P'), displayed via GRID->SET_TOP_OF_LIST or footer aggregation (LVC_S_LAYO-TOTALS_TEXT).

```
*&---------------------------------------------------------------------*
*& Report ZHMS_R07
*&---------------------------------------------------------------------*
*&
*&---------------------------------------------------------------------*
REPORT zhms_r07.

TABLES : zhms_patient, zhms_bill_hdr, zhms_bill_item.

TYPE-POOLS : slis.

*--- Header Structure
TYPES : BEGIN OF str_head,
        patient_id     TYPE zhms_bill_hdr-patient_id,
        bill_no        TYPE zhms_bill_hdr-bill_no,
        bill_date      TYPE zhms_bill_hdr-bill_date,
        total_amount   TYPE zhms_bill_hdr-total_amount,
        payment_status TYPE zhms_bill_hdr-payment_status,
        rowcolor       TYPE c LENGTH 4,   "<<< row color
       END OF str_head.

*--- Item Structure
TYPES : BEGIN OF str_item,
        bill_no   TYPE zhms_bill_item-bill_no,  "<<< link key
        item_no   TYPE zhms_bill_item-item_no,
        quantity  TYPE zhms_bill_item-quantity,
        rate      TYPE zhms_bill_item-rate,
        amount    TYPE zhms_bill_item-amount,
       END OF str_item.

DATA : it_head TYPE TABLE OF str_head.
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DATA : wa_head TYPE str_head.
DATA : it_item TYPE TABLE OF str_item.
DATA : wa_item TYPE str_item.

*--- ALV
DATA : it_fcat    TYPE slis_t_fieldcat_alv.
DATA : wa_fcat    TYPE slis_fieldcat_alv.
DATA : wa_layout  TYPE slis_layout_alv.
DATA : wa_keyinfo TYPE slis_keyinfo_alv.

*--- Footer totals
DATA : lv_collected   TYPE zhms_bill_hdr-total_amount.
DATA : lv_outstanding TYPE zhms_bill_hdr-total_amount.
DATA : it_listheader  TYPE slis_t_listheader.
DATA : wa_listheader  TYPE slis_listheader.


SELECTION-SCREEN : BEGIN OF BLOCK b1 WITH FRAME TITLE TEXT-000.
  SELECT-OPTIONS : s_pat    FOR zhms_bill_hdr-patient_id.
  SELECT-OPTIONS : s_status FOR zhms_bill_hdr-payment_status.
  SELECT-OPTIONS : s_date   FOR zhms_bill_hdr-bill_date.
SELECTION-SCREEN : END OF BLOCK b1.


START-OF-SELECTION.

  PERFORM fetch_data.
  PERFORM calc_totals.
  PERFORM build_fcat.
  PERFORM display_alv.


*&---------------------------------------------------------------------*
*& FORM FETCH_DATA
*&---------------------------------------------------------------------*
FORM fetch_data.

*--- Fetch HEAD
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```
 SELECT patient_id,
        bill_no,
        bill_date,
        total_amount,
        payment_status
   INTO TABLE @it_head
   FROM zhms_bill_hdr
   WHERE patient_id     IN @s_pat
     AND payment_status IN @s_status
     AND bill_date      IN @s_date.

 IF it_head IS INITIAL.
   MESSAGE 'No billing records found' TYPE 'I'.
   LEAVE LIST-PROCESSING.
 ENDIF.

*--- Fill row color based on payment status
 LOOP AT it_head INTO wa_head.
   CASE wa_head-payment_status.
     WHEN 'P'.
       wa_head-rowcolor = 'C510'.   "Green  - Paid
     WHEN 'n'.
       wa_head-rowcolor = 'C610'.   "Red    - Outstanding
     WHEN OTHERS.
       wa_head-rowcolor = 'C310'.   "Yellow - Others
   ENDCASE.
   MODIFY it_head FROM wa_head.
 ENDLOOP.

*--- Fetch ITEM using FOR ALL ENTRIES
 SELECT bill_no,
        item_no,
        quantity,
        rate,
        amount
   INTO TABLE @it_item
   FROM zhms_bill_item
   FOR ALL ENTRIES IN @it_head
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   WHERE bill_no = @it_head-bill_no.

  SORT it_item BY bill_no ASCENDING
                  item_no  ASCENDING.

ENDFORM.


*&---------------------------------------------------------------------*
*& FORM CALC_TOTALS
*& Calculate Total Collection and Outstanding
*&---------------------------------------------------------------------*
FORM calc_totals.

  CLEAR : lv_collected, lv_outstanding.

  LOOP AT it_head INTO wa_head.
    IF wa_head-payment_status = 'p'.
      lv_collected = lv_collected + wa_head-total_amount.
    ELSE.
      lv_outstanding = lv_outstanding + wa_head-total_amount.
    ENDIF.
  ENDLOOP.

ENDFORM.


*&---------------------------------------------------------------------*
*& FORM BUILD_FCAT
*& ONE combined field catalog — tabname distinguishes HEAD vs ITEM
*&---------------------------------------------------------------------*
FORM build_fcat.

*--- HEAD columns
  CLEAR wa_fcat.
  wa_fcat-col_pos   = 1.
  wa_fcat-tabname   = 'IT_HEAD'.
  wa_fcat-fieldname = 'PATIENT_ID'.
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```
  wa_fcat-seltext_m = 'Patient ID'.
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 2.
  wa_fcat-tabname   = 'IT_HEAD'.
  wa_fcat-fieldname = 'BILL_NO'.
  wa_fcat-seltext_m = 'Bill No'.
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 3.
  wa_fcat-tabname   = 'IT_HEAD'.
  wa_fcat-fieldname = 'BILL_DATE'.
  wa_fcat-seltext_m = 'Bill Date'.
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 4.
  wa_fcat-tabname   = 'IT_HEAD'.
  wa_fcat-fieldname = 'TOTAL_AMOUNT'.
  wa_fcat-seltext_m = 'Total Amt'.
  wa_fcat-do_sum    = 'X'.          "<<< Grand total
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 5.
  wa_fcat-tabname   = 'IT_HEAD'.
  wa_fcat-fieldname = 'PAYMENT_STATUS'.
  wa_fcat-seltext_m = 'Pay Status'.
  APPEND wa_fcat TO it_fcat.

*--- ITEM columns
  CLEAR wa_fcat.
  wa_fcat-col_pos   = 1.
  wa_fcat-tabname   = 'IT_ITEM'.
  wa_fcat-fieldname = 'ITEM_NO'.
  wa_fcat-seltext_m = 'Item No'.
```

```
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 2.
  wa_fcat-tabname   = 'IT_ITEM'.
  wa_fcat-fieldname = 'QUANTITY'.
  wa_fcat-seltext_m = 'Quantity'.
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 3.
  wa_fcat-tabname   = 'IT_ITEM'.
  wa_fcat-fieldname = 'RATE'.
  wa_fcat-seltext_m = 'Rate'.
  APPEND wa_fcat TO it_fcat.

  CLEAR wa_fcat.
  wa_fcat-col_pos   = 4.
  wa_fcat-tabname   = 'IT_ITEM'.
  wa_fcat-fieldname = 'AMOUNT'.
  wa_fcat-seltext_m = 'Amount'.
  wa_fcat-do_sum    = 'X'.          "<<< Item subtotal
  APPEND wa_fcat TO it_fcat.

ENDFORM.


*&---------------------------------------------------------------------*
*& FORM BUILD_FOOTER
*&---------------------------------------------------------------------*
FORM build_footer.

  DATA : lv_col_txt TYPE char30.
  DATA : lv_out_txt TYPE char30.

  WRITE lv_collected   TO lv_col_txt.
  WRITE lv_outstanding TO lv_out_txt.
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```
  CLEAR wa_listheader.
  wa_listheader-typ  = 'H'.
  wa_listheader-info = 'Billing Summary'.
  APPEND wa_listheader TO it_listheader.

  CLEAR wa_listheader.
  wa_listheader-typ  = 'S'.
  wa_listheader-key  = 'Total Collection  :'.
  wa_listheader-info = lv_col_txt.
  APPEND wa_listheader TO it_listheader.

  CLEAR wa_listheader.
  wa_listheader-typ  = 'S'.
  wa_listheader-key  = 'Outstanding Amt   :'.
  wa_listheader-info = lv_out_txt.
  APPEND wa_listheader TO it_listheader.

ENDFORM.


*&---------------------------------------------------------------------*
*& FORM DISPLAY_ALV
*&---------------------------------------------------------------------*
FORM display_alv.

  PERFORM build_footer.

* Layout
  wa_layout-info_fieldname    = 'ROWCOLOR'.
  wa_layout-colwidth_optimize = 'X'.
  wa_layout-totals_text       = 'Grand Total :'.

* Key
  wa_keyinfo-header01 = 'BILL_NO'.
  wa_keyinfo-item01   = 'BILL_NO'.

*--- Register END_OF_LIST event for footer
  DATA : it_events TYPE slis_t_event.
```

```
 DATA : wa_event  TYPE slis_alv_event.

 wa_event-name    = slis_ev_end_of_list.  "<<< footer event
 wa_event-form    = 'SHOW_FOOTER'.        "<<< FORM to call
 APPEND wa_event TO it_events.

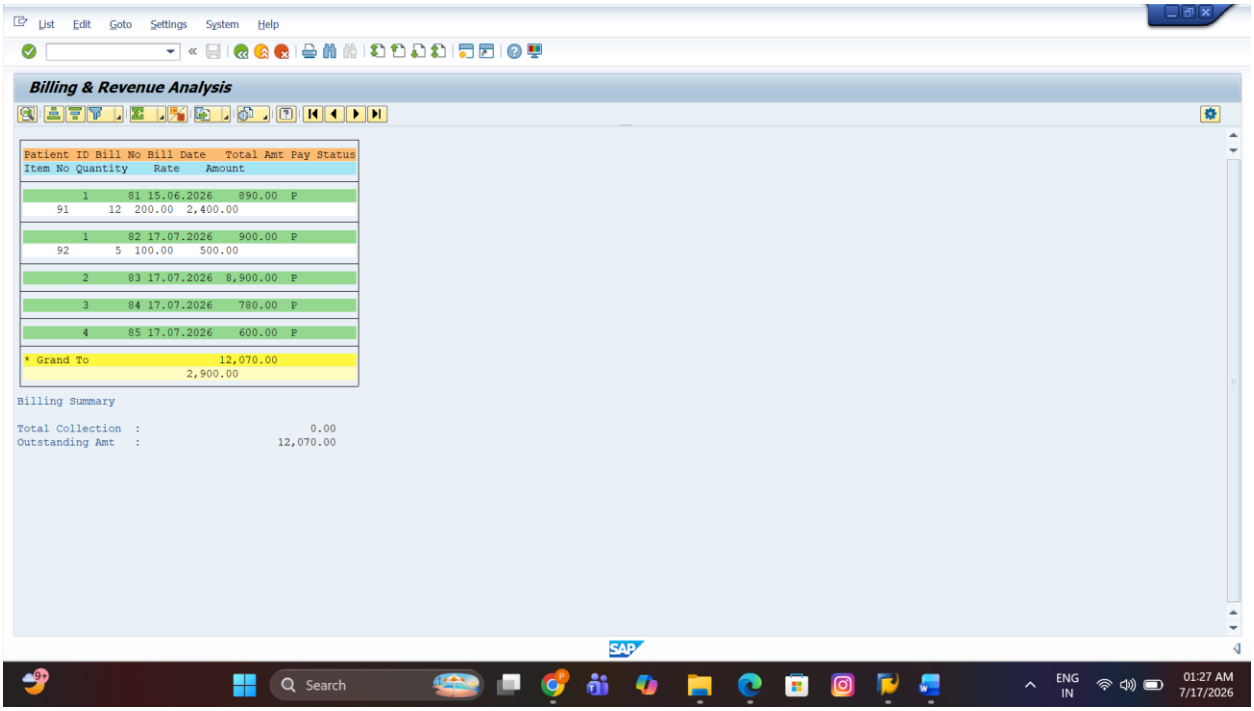
 CALL FUNCTION 'REUSE_ALV_HIERSEQ_LIST_DISPLAY'
  EXPORTING
   i_callback_program = sy-repid
   is_layout         = wa_layout
   it_fieldcat       = it_fcat
   i_tabname_header   = 'IT_HEAD'
   i_tabname_item     = 'IT_ITEM'
   is_keyinfo        = wa_keyinfo
   it_events         = it_events       "<<< CORRECT parameter
  TABLES
   t_outtab_header    = it_head
   t_outtab_item      = it_item
  EXCEPTIONS
   program_error      = 1
   OTHERS             = 2.

 IF sy-subrc <> 0.
  MESSAGE 'ALV Display Error' TYPE 'E'.
 ENDIF.

ENDFORM.


*&---------------------------------------------------------------------*
*& FORM SHOW_FOOTER — called by END_OF_LIST event
*&---------------------------------------------------------------------*
FORM show_footer.

 CALL FUNCTION 'REUSE_ALV_COMMENTARY_WRITE'
  EXPORTING
   it_list_commentary = it_listheader.  "<<< footer data
ENDFORM.
```


List Edit Goto Settings System Help
Billing & Revenue Analysis
Patient ID Bill No Bill Date Total Amt Pay Status
Item No Quantity Rate Amount
1 81 15.06.2026 890.00 P
91 12 200.00 2,400.00
1 82 17.07.2026 900.00 P
92 5 100.00 500.00
2 83 17.07.2026 8,900.00 P
3 84 17.07.2026 780.00 P
4 85 17.07.2026 600.00 P
* Grand To 12,070.00
2,900.00
Billing Summary
Total Collection : 0.00
Outstanding Amt : 12,070.00
SAP
Search
ENG
IN
01:27 AM
7/17/2026